Does Rapid Transit and Light Rail Infrastructure Improve Labor Market Outcomes?

by

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Abstract

Public transit has often been proposed as a solution to the spatial mismatch hypothesis but the link between public transit accessibility and employment has not been firmly established in the literature. Los Angeles provides an interesting case study – as the city has transformed from zero rail infrastructure before the 1990s to a large network consisting of subway, light rail, and bus rapid transit servicing diverse neighborhoods. I use confidential panel data from the American Community Survey, treating route placement as endogenous, which is then instrumented by the distance from the centroid of each tract in LA to a hypothetical Metro route. Overall, I find proximity to Metro stations increases employment for residents, which is robust to using both a binary and continuous measure of distance. Additionally, I find evidence that increased job density in neighborhoods near new transit stations is contributing to the employment increase.

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1. Introduction:

1.1 Motivation

I evaluate the effects of expanding the Los Angeles Metro rapid transit network of light rails, subways, and bus rapid transit (BRT)[1] on labor force participation and employment in the neighborhoods in which these stations are placed. Los Angeles has been consistently expanding its Metro network since the initial construction of a light rail line in 1990, following the absence of the last consumer streetcar in 1963, to the completion of 103 Metro stations by 2016. This paper is primarily motivated by the fact that public transit has often been proposed as a solution to "spatial mismatch", historically defined by economists as a mismatch between where low-income households reside and suitable job opportunities. Spatial mismatch traditionally focuses on the effects of residential segregation in the urban core for low-skilled and minority workers and the suburbanization of job opportunities far away from the urban core (Kain, 1968; Kain, 1992; Gobillon et al., 2007). While researchers and policymakers have often proposed public transportation improvements to address spatial mismatch, it remains unclear whether public transit improvements have a significant effect on the labor market outcomes.

The papers that address the link between public transit and employment are limited and generally focus on accessibility of transit in terms of cost (Phillips, 2014; Rodriguez et al. 2015), racial hiring decisions of employers as a result of increased transit (Holzer et al. 2003) or have weak identification strategies that do not fully address endogenous placement of routes (Sanchez 1999; Sari 2015; Heilmann 2018). This paper provides several contributions. First, it expands on a sparse literature tying public transit infrastructure to employment outcomes and addresses endogenous placement of routes that has not been addressed in previous studies. Additionally, from a public policy perspective, the infrastructure expansion was costly and voted directly by referendum to increase gas taxes in Los Angeles County. By evaluating the effects on the labor market, this paper addresses whether there were tangible benefits to

[1] The decision to include the Orange line, a bus-rapid transit system, is because of its dedicated right-of-way and an on-time performance of 94%, close to the 99% of the train system (Mendelson 2015)

funding Metro expansions. Finally, it addresses the spatial mismatch component by observing whether transit accessibility can increase employment and reduce the disparities in labor market outcomes among different subgroups more vulnerable to this mismatch.

Using Los Angeles as a case study also addresses the spatial mismatch hypothesis better than other cities, because the high concentration of minority groups and large variations in income across neighborhoods allows the evaluation of the differential employment effects across racial and low-income groups. Additionally, the Los Angeles County Metropolitan Transportation Authority (MTA) is attempting to finish construction of a series of expansions by the 2028 Summer Olympics. The effects of long-run infrastructure improvements on labor market conditions resulting from sporting events, such as the expansion of public transit, has not been fully explored in the literature, which tends to focus on the short-run effects of sporting events on employment.[2] Much of the literature on public transit infrastructure improvements focuses on how Metro stations have a gentrification effect by increasing housing prices. (Kahn 2007; Bowes and Ihlanfeldt 2001; Zheng and Kahn 2013; Hu 2017; Dziauddin 2019; Zhou et al. 2020). However, there's little work on how the impact of public transit infrastructure can increase commercial activity and sprout new businesses in the areas, thereby improving the employment opportunities for their residents. While one may think of public transit increasing employment by connecting residents to job opportunities that were otherwise inaccessible due to commuting costs, it can also increase job density in the neighborhood and reduce spatial mismatch.

To briefly summarize the method and results of the paper, I first evaluate the effects of proximity to a station on public transit usage, labor force participation, and employment using a standard panel data model with tract and time fixed effects. However, route placement may be endogenous to changes in existing demand for public transportation and neighborhood composition. Thus, I calculate the distance of the centroid of each tract to a hypothetical Metro route as an instrument to predict new station locations.

[2] A large consensus seems to be that the positive employment effects are minimal, at best, and often fall short of initial projections or statistics published by governmental sources (Baade and Matheson, 2002; Hotchkiss et al., 2003; Baumann et al., 2012).

The hypothetical Metro route is modelled from a least-cost route that minimizes elevational slope changes to connect the origin and destinations of Metro stations. Overall, I find that proximity to Metro stations increases labor force participation and employment, reducing spatial mismatch. These positive effects are typically higher for the instrumental variables (IV) approach relative to the two-way-fixed effects (TWFE) approach. Additionally, the results are consistent regardless of whether I use a binary or continuous measure of proximity to Metro stations. Finally, I find that the increased employment is likely driven by increased job densities in neighborhoods close to Metro stations rather than allowing residents to connect to distant job opportunities.

1.2. Brief History of LA Metro:

In the early 1900s, Los Angeles had an extensive urban electric streetcar network primarily owned by two companies, Pacific Electric and the Los Angeles Railway, colloquially known as the red cars and yellow cars. However, due to the proliferation of automobiles leading to congestion, the unprofitable fare restrictions and maintenance, and allegations of interference from the automobile industry to switch to buses, all streetcar systems closed by 1963 (Marshall 2016).[3]

In 1975, there were initial talks of constructing a modern rapid transit system in Los Angeles. For the next few decades, planners identified areas that they wished to connect by transit, but a series of political, legal, and logistical hurdles significantly delayed construction. The first route, the Blue line, which had been proposed since 1975, finally became operational in 1990. Soon after, the Red/Purple line was constructed in two stages, in 1993 and 1996 due to some issues with route placement that will be discussed later, and the Green line was constructed in 1995. However, despite the acquisition of a wide network of 175 miles of former Pacific Electric rights-of-way[4] in 1990, development soon fizzled out due

[3] See Figure 1 for a timeline of relevant events.
[4] A right-of-way is a type of easement granted or reserved over the land for transportation purposes.

to budgetary issues, the economic downtown in the early 1990s, and the decreased public standing of the Metropolitan Transit Authority (MTA). More importantly, residents in the areas initially mobilized heavy resistance against transit routes due to concerns about neighborhood safety.

However, in the early 2000s, there was a shift in public opinion back towards developing rapid transit due to increased traffic, an increase in federal and state grants, and the increase in politicians favorable to transit (Elkin 2014). This led to the development of the Gold line in 2003 that connected Union Station to Pasadena, the Orange bus rapid transit system (BRT) in 2005 from Hollywood through the San Fernando Valley, the Gold line light rail expansion from Union Station to East Los Angeles in 2009, an expansion to the Orange line BRT in 2012, the first leg of the Expo light rail construction in 2012 that connects downtown to Culver City, the expansion of the Expo line in 2016 that connects to Santa Monica, and the expansion of the Gold line from Pasadena to Azusa in 2016. Additionally, in 2008, the passage of a proposition, Measure R, secured $40 billion dollars for MTA over thirty years, with 35% dedicated to transit capital projects. This measure helped guarantee the future for its current and planned projects through 2026. Figure 2 shows all routes in 2000, 2016, and the planned, future routes through 2026 (L.A. Metro).

2. Literature Review:

2.1 Spatial Mismatch

I divide the literature review into two sections, one introducing the concept of spatial mismatch and the other on the current literature regarding public transit and employment. Before spatial mismatch was first introduced, Mills described a model of urban residential choice that assumes individuals choose optimal residential locations based on employment in a fixed city center and face tradeoffs between housing costs and commute times based on their choice of residency (Mills 1972). However, due to the decentralization of employment and individual preferences for amenities, there may be a mixture of high

and low-income individuals who reside in the central city or the suburban areas (White 1988) and residents living in the suburbs generally have higher incomes to compensate for longer travel times (Madden 1985).

However, many people face barriers to obtaining their preferred residential locations. Kain (1968) explained the high unemployment rate for Blacks as stemming from housing segregation and the suburbanization of job opportunities. Kain's paper focused on Chicago and Detroit and how low-skilled jobs moved from the urban core of cities to the suburbs, while Black workers were still constrained to the urban core due to housing market discrimination. This idea was later dubbed "spatial mismatch", a disconnect between where low-income residents reside and suitable job opportunities, and a large subsequent literature stressed the importance of proximity to workplace or accessibility of jobs on employment.[5]

Gobillon et al. (2007) suggested seven mechanisms on why greater distances to job opportunities could be harmful. On the worker's side, four possible explanations are that workers may refuse a job that involves a long and costly commute, workers' job search efficiencies may decrease with distance to jobs, workers residing far away from jobs may not search intensively, and workers may incur high search costs that restrict their spatial search horizon to nearby neighborhoods. The first mechanism can be explained by higher reservation wages for costly commutes, resulting in lower employment for individuals far away from job opportunities (Brueckner and Zenou 2003; Coulson et al. 2001). Additionally, barriers to search exist from the reduced availability of job information (Ihlanfeldt and Sjoquist 1990, Ihlanfeldt and Sjoquist 1991) and workers may be deterred from high search costs and restrict their search to nearby neighborhoods (Stoll 2005, Holzer and Reaser 2000). Under these labor supply side explanations, improvements in reliable public transit methods should decrease commuting costs and increase search intensity, resulting in a positive effect on employment.

[5] For a detailed literature review on spatial mismatch papers, see Gobillon et al. (2007) and Holzer (1991)

On the labor demand side, Gobillon et al. (2007) also suggested three firm-related mechanisms for spatial mismatch. Employers may discriminate against residentially segregated workers because of stigma or prejudice associated with their residential location (statistical discrimination), employers may refuse to hire or prefer to pay lower wages to distant workers because of the fear that long commutes will make a worker less productive, and suburban employers may think their White local customers are unwilling to have contact with minority workers (customer discrimination). There is some evidence that firm-based discrimination may occur. Hellerstein et al. (2008) suggested the problem is less of a spatial mismatch but rather a "racial mismatch", where there is a lack of jobs where minority groups are hired into. Phillips (2014) runs a correspondence study and finds employers discriminate against employees who live further from job locations. Under the first and third scenarios that have a strong racial discrimination component, improvements in public transit will likely have a null effect on employment, as transit accessibility will not necessarily address these problems.[6] The second scenario regarding the employer's fear of decreased worker productivity is a bit unclear; public transportation can reduce commuting costs which should increase the perception of worker productivity.

2.2 Public Transportation as a Solution to Spatial Mismatch

Spatial mismatch papers often recommend improvements in public transit infrastructure and accessibility as a potential solution to addressing the issue (Kain 1968). However, despite this insistence on public transit infrastructure improvements as a viable solution, the evidence in the literature addressing the connection between public transit infrastructure and employment is somewhat limited in their identification strategies.[7] For example, while a few early studies find a positive association between

[6] There may be an exception if public transit improvements increase job density for jobs that minorities can be hired into, reducing the racial mismatch component.

[7] There is a growing literature focused on the effects of reducing transit accessibility in terms of cost on employment outcomes using randomized controlled trials (see Phillips 2014; Rodriguez et al 2015). My study focuses on transit accessibility in terms of the availability of infrastructure rather than the cost of using public transit.

public transit and employment at the intensive margin (Sanchez 1999; Kawabata 2003), none of these studies sufficiently address endogeneity issues, relying on regression control and cross-sectional data. Holzer et al. (2003) find that firms located near transit stations in San Francisco hired more Latino workers, but this type of analysis focuses on the decision-making process of firms rather than the benefits to residents of an expanded transit network.

More recently, there have been papers that use difference-in-differences or two-way fixed effects models to address the link between public transit and employment (Sari 2015; Heilmann 2018).[8] Sari (2015) finds a new tramway in Bordeaux, France decreased unemployment rates in neighborhoods located close to the tramway station more relative to other neighborhoods. For an example from the United States, Heilmann (2018) find positive effects on income from the construction of Dallas's urban rail network. However, the methodology could be problematic if station placement is correlated with changes in existing demand for public transit infrastructure. In Heilmann (2018) for the case of Dallas's train, the initial plan was to construct over 160 miles of rail, which was later curtailed to 93 miles of rail. Thus, it is possible that Dallas planners may have chosen which plans to curtail depending on the changes in economic conditions, ridership, or transportation demands in specific neighborhoods. In my study, I address any potential biases from Metro line placement decisions with instrumental variables, an approach that has not been used in these previous studies. Severan (2019) studied welfare effects resulting from the construction of the LA Metro system. However, there are a few notable differences. First, Severan focuses on different time periods and uses 1990 and 2000 census data to evaluate the first few routes constructed in the 1990s. Secondly, Severan builds a spatial general equilibrium model that accounts for commuting, housing markets, and labor markets to determine the overall welfare effect. While both Severan and this paper study the same setting, ultimately, they take different approaches in both their estimation strategy and outcome of interest. My paper's goal is to address the spatial mismatch

[8] Aslund et al (2017) find a zero impact of commuter train access in Sweden on earnings and employment. However, intercity commuter trains are fundamentally different from rapid intracity transportation systems, and I do not explore similar systems such as the Metrolink in this paper, mainly due to data limitations.

question by estimating the average employment and earnings effect for residents and whether this is attributed to increased job density nearby.

Finally, while transit infrastructure improvements may address spatial mismatch by connecting residents to farther-away jobs that were otherwise inaccessible, thereby increasing overall employment, residents may also increase their employment from increased job opportunities in areas near transit stations. However, the literature on that is unclear. Credit (2017) finds growth in new businesses in the knowledge, service, and retail sectors after station openings from the Phoenix light rail system, although the effect tapered off over time. Schuetz (2015) do not find positive effects of Metro stations on retail activity in four California MSAs, including Los Angeles from 1992-2009. The follow-up to this paper suggests that the retail growth may be lagged 5-10 years (Schuetz et al. 2018). However, there is a larger and more consistent literature on the gentrification effects of public transportation improvements. Overall, these papers find that proximity to transit stops increases property values or housing in the area (Kahn 2007; Bowes and Ihlanfeldt 2001; Hu 2017; Dziauddin 2019; Zhou et al. 2020).

3. Data:

I mainly use confidential 1-year data from the American Community Survey (ACS) ranging from 2006 to 2018.[9] This allows us to evaluate the effects of public transit infrastructure expansions in 2009, 2012, and 2016. The yearly data from 2006 to 2009 is converted from 2000 census tract boundaries to 2010 census tract boundaries using the Longitudinal Tract Database (LTDB).[10] I measure the effects on employment to population ratio, population, subway or rail usage, earnings, commuting time, number of jobs, and average earnings from those jobs. Additionally, I retrieve demographic data such as race, gender, education, migration, and vehicle ownership as controls. The first iteration of this paper used 5-

[9] An earlier draft of this paper was done using public 5-year ACS data.
[10] LTDB uses an interpolation based on both geographical area and population weights. For a discussion of the benefits of LTDB over the Neighborhood Change Database (NCDB), see Logan et al. (2014). Lee and Lin (2018) also prefer LTDB over NCDB for converting 2000 tracts into 2010 tracts.

year public data on from the ACS, and a brief summary of the methodology and results from the analyses using the 5-year data can be found in Appendix A.

I also use the Longitudinal Employer-Household Dynamics Origin-Destination Employment Statistics (LODES) data to measure the effects of proximity to Metro stations on job density as a supplementary analysis. LODES provide the total number of jobs in each origin-destination pair. In the workplace dataset, the number of jobs is provided as well as more detailed breakdowns by race and gender, which we use to compare with the ACS results. Graham et al. (2014) addresses some of the differences in approach between ACS and LODES. The most relevant differences are that LODES data reports place of work based on federally administrative data rather than a physical address reported by the individual surveyed. Additionally, in handling small sample sizes, LODES employs various noise infusion methods and small sample imputation methods while ACS typically suppresses output. The latter proves to be problematic for generating a large enough sample size to use the ACS for measuring job density.

Finally, I retrieve information about the location and timing of Metro stations from the LA Metro website. Addresses are geocoded into ArcMap and overlaid on top of census tract shape files. ArcMap allows a variety of spatial analysis tools, such as the calculation of distances from a point or a line to the centroid of the tract and the calculation of least-cost paths, which will be explained in the econometric specifications.

4. Estimation:

4.1 Two-way Fixed Effects

The model in equation (1) represents a simple two-way fixed effects regression with tract and year dummy variables that allows for the pooling of multiple Metro expansions and variation in the data across many years.

$$Y_{ct} = \alpha + \beta_1 * Metro_{ct} + \beta_2 * X_{ct} + \Upsilon_c + \lambda_t + \varepsilon_{ct} \quad (1)$$

In equation (1), $Metro_{ct}$ indexes whether a tract *c* contains a Metro station at year *t* within walking distance. Walking distance has traditionally been defined in the urban planning literature as 0.4 kilometers from transit stops although some studies suggest that rail transit may be higher than 0.8 kilometers.[11] For the purpose of this study on rapid transit systems, I use a 0.8 km buffer around the Metro station.[12]

Y_{ct} indexes the outcomes of interest. I measure the average effect of a tract in LA having a Metro station within walking distance on residential employment, population, subway or rail usage, earnings within the past twelve months, commuting time, number of jobs, and average earnings from those jobs from the past twelve months. The controls in X_{ct} include tract level shares of Black, female, high school degree, some college, bachelor's or higher, portion of the tract population that do not have vehicles, and proportion of the tract population with only one vehicle. We exclude all residents who have moved in the past year to address a potential neighborhood sorting endogeneity concern that people who are already employed migrate to tracts near the stations after they are constructed. λ_t indicates a vector of dummy variables for year and Υ_c indicates a vector of dummy variables for census tracts. Because I use tract-level averages to construct the panel dataset, I weight the regression by the average population of the studied group when appropriate (i.e., total population (16+) when analyzing employment for the total population, Black population when analyzing employment for Blacks, etc.).[13] While the mean population size of a tract is around 3000, there are a couple of tracts with very small population sizes (see Figure 3).[14] Therefore, I removed all tracts under 100 population.

There are potential shortcomings with using a binary treatment variable. Firstly, if a tract boundary touches the buffer zone, it will be considered a treated tract. Due to the somewhat irregular shapes of some of the tracts, it is unlikely that the entire tract is within the walking distance buffer – and

[11] For a detailed literature review on the usage of different walking distance buffers to transit, see El-Geneidy et al. (2014)
[12] Using a smaller 0.4 km buffer yielded similar results.
[13] I test out each specification without the population weights as well. In the end, the resulting estimates do not seem to differ much. Results for unweighted regressions are available upon request.
[14] This is based off public ACS data.

the differences in the geographical sizes of the tract can be a source of measurement error. More importantly, residents may have different propensities to walk longer distances whereas a fixed walking distance buffer may not be appropriate. For example, higher household incomes and vehicle availability negatively affect a person's propensity to walk (Hsiao et al. 1997). To address this issue, I also employ a continuous distance-based treatment variable that measures the effect of proximity to a Metro transit station on employment in equation (2).

$$Y_{ct} = \alpha + \beta_1 * \text{Proximity}_{ct} + \beta_2 * X_{ct} + \Upsilon_c + \lambda_t + \varepsilon_{ct} \quad (2)$$

Proximity$_{ct}$ indexes the negative distance in kilometers from the centroid of the tract *c* to the nearest Metro station in time *t*.[15] As new stations are built, tracts that are near the new station will increase their proximity to the Metro station (in other words, the distance in kilometers decreases). If a tract's closest station is unaffected by the new Metro station expansion, then *Proximity*$_{ct}$ will remain unchanged. By using a continuous rather than a discrete measure for Metro stations, I address the issue of residents with different propensities to walk. I express *Proximity*$_{ct}$ using the inverse arcsine to allow for a greater emphasis on changes at smaller distances from a new Metro station. It is important to not use levels, because it is not realistic to expect a constant effect from increasing proximity; a one-kilometer increase in proximity due to a new station has a larger effect in tracts close to Metro stations whereas a one-kilometer increase in proximity from tracts far from the station is not expected to have the same effect. I also express Y_{ct} using the inverse arcsine for easier interpretability.[16] [17]

4.2 IV

[15] If a tract is 2 kilometers away from the Metro station, the value for proximity for that tract will be -2.
[16] Inverse arcsine is used to approximate a percentage change in place of natural logs because of the presence of tracts with zeros in the outcome variables.
[17] I also test other functional forms, such as levels or 1/KmFromStation. However, since the levels regressions are prone to being affected by outliers and do not account for decreasing distances and the reciprocal form does not have a clean interpretation, the regressions using inverse arcsine are preferred.

The construction of a new Metro line may not be exogenous, and may be dependent on a variety of factors, such as existing demand for public transportation, economic indicators, political factors, and geographical practicality. While tract fixed effects assist in controlling for unobserved time-invariant factors, some factors determining Metro placement may vary over time. For example, if there are large changes in existing demand for transportation in a tract, Metro planners may purposely target these areas in their expansion plans. Therefore, using an instrumental variables specification can reduce the bias attributed to endogenous determinants of route placement.

Elkin (2014) provides a history of the decision-making process in rail lines and motivates the creation of an appropriate instrument.[18] As shown in Figure 1's timeline, oftentimes routes have pre-determined origin and destination areas that are determined well in advance. The idea of starting with a connection from downtown L.A to Long Beach largely originated from a political compromise in November 21, 1975 between Mayor Bradley and local supervisor Baxter Ward, well before the first construction of the Blue line in 1990.[19] [20] Kenny Hahn, a county supervisor and chair of the Los Angeles County Transportation Commission (LACTC),[21] successfully pushed a proposition for a sales tax hike to fund transit rail plans. Largely due to his individual efforts, the downtown to Long Beach route was prioritized and chosen as the first route in March 1982, which will pass through his predominantly Black, transit-dependent district (Elkin 2014). However, even though the destination was known, this route had resistance to what path it should take. Some wanted a heavy rail route down Vermont Avenue, a busier transit corridor that could service more residents. However, acquiring the funding and political backing

[18] Elkin (2014) in the book *Railtown: The Fight for the Los Angeles Metro Railway and the Future of the City* provides a succinct historical account of the decision-making process in transit. His primary focus is how the political decisions of rail advocates and detractors and the public support of neighborhoods was the critical factor in determining which routes were prioritized and where they ended up going.

[19] Bradley wished to create a downtown and Hollywood-based subway via Wilshire Blvd, Ward wanted a regional above-ground rail network using existing right of ways (suburban-oriented) (Elkin 2014). These will later develop into the Red/Purple and Blue lines.

[20] After this decision, the federal government organization, the Urban Mass Transportation Administration (UMTA) did not provide funds to the original organization RTD until June 1980 and they insisted on using the funds for a subway system along the Wilshire corridor, closer to Bradley's vision, which will form the basis for the Red Line. A newly-formed organization, LACTC, will secure funding for the Blue Line.

[21] The LACTC and the Southern California Rapid Transit District (RTD) combine into the MTA in 1993

would have been more difficult over building on an existing right-of-way owned by Pacific Electric.[22] Even after finally deciding on following the Pacific Electric right-of-way over the Vermont Avenue route, the LACTC still struggled with the determination of the paths within downtown Los Angeles and downtown Long Beach which were not part of the acquired right-of-way. The plans for the actual route were not finalized until March 1985, nearly a decade after the route was first conceived.

Similarly, for the future Red Line, the Southern California Rapid Transit District (RTD) initially decided in September 1979 on a plan to connect to Hollywood via a route that first goes west to Wilshire and then goes back North and East to Hollywood. However, after a methane gas explosion in 1985, Henry Waxman, the U.S. Representative for the 33rd congressional district, opposed the subway construction through the Wilshire corridor. His concerns were about the risk of another methane gas explosion from the construction; however, he might have also been influenced by the mobilization of his constituency (the wealthy residents of Hancock Park, Beverly Hills, and Fairfax) against the route due to concerns about noise, commuters, and traffic. Therefore, the route had to be changed, and planners decided on an alternative route up through the Vermont corridor to Hollywood and only created a small route down Wilshire, as far as Waxman allowed the route to go.[23]

The bottom line is that from as early as 1981-1984, the transportation authorities identified areas where they wished for the routes to be built, including connections from downtown to the San Fernando Valley, Santa Monica, East Los Angeles, and Pasadena (Elkin 2014).[24] Figure 4 shows a few of these early proposals and clearly identifies the end destinations of most of the current and future Metro routes. Origins and destination areas are known relatively far in advance, which I treat as exogenous, but the path chosen to connect these areas can be endogenous to factors such as community support and neighborhood demand for transit. To address this endogeneity, I use an instrument to predict where Metro stations are

[22] Pacific Electric owned most of the rights-of-way of their former trolley system, the Red Cars.
[23] The Vermont corridor was chosen in part due to limited residents in the area and its composition as more of a business corridor, with a large number of hospitals.
[24] San Fernando Valley line will later become the Orange Line, the connection to Santa Monica will be the Expo line, and the Pasadena & East LA routes will become the Gold Line

built. This instrument will measure the distance of the centroid of each tract to the nearest point on a hypothetical Metro route connecting the origin and destinations of a specific expansion. Therefore, I am addressing endogeneity from route placement rather than the exact station location along a given route.

I construct a hypothetical Metro route using a least-cost hypothetical route that connects the origin and planned destination by minimizing the changes in the slope of elevation. I constructed this route using a 10 ft digital elevation model (DEM) of Los Angeles County and converted the DEM to a slope raster, which calculates the rate of change in elevation from each cell in the map. The least-cost path is determined by calculating a cost path that minimizes the cost distance, which uses the previous slope raster as the input. Figure 5 shows the hypothetical routes that connects the origin and destination points for the Gold line expansion in 2009, phase 1 of the Expo line in 2012, and the expansion of the Orange line in 2012. From this figure, it should be clear that as the distance from any tract *c* and the least-cost hypothetical route increases, the probability of a Metro station being constructed in tract *c* decreases. Conversely, when using the continuous treatment variable, the kilometers from the nearest Metro station, as the distance to the hypothetical route increases, so should the distance to the nearest actual Metro station. There are a few usages of constructing least-cost hypotheticals to use as instruments. Lipscom et al. (2013) and Kaseem (2018) construct a hypothetical electrical grid in the countries they study based on geographical-based cost considerations. Neumark (2008) uses distance to Benton County to predict the opening of new Walmart stores. The benefit of this type of instrument is that along with a strong first stage, the theoretical argument for the exclusion principle is strong. It is unlikely that the distance of the centroid of each tract to a hypothetical Metro route affects labor market outcomes except through its prediction of actual stations.

More formally, the first-stage IV specification relating to eq (1) is as follows:

$$Metro_{ct} = \alpha + \delta * Hypo_Route_c * post_t + \lambda_2 * X_{ct} + \Upsilon_c + \lambda_t + \nu_{ct} \quad (3)$$

$Hypo_Route_c$ is a vector of distance dummies that indicate whether the distance from the centroid of tract c is less than 1km, 2km, or 3km from the tract centroid to the hypothetical route.[25] $Post_t$ is an indicator for the post-treatment period for 2009 through 2018, following the first expansion in the analysis period. This is necessary because $Hypo_Route_c$ does not vary with time, so by interacting it with $post_t$ allows the instrument to not be absorbed by the fixed effects. Additionally, the distance to the hypothetical route is not relevant for new Metro stations until the construction period. An equivalent distance-based IV first-stage modification of equation (3) would be as follows:

$$Proximity_{ct} = \alpha + \delta_1 * Hypo_Route_Dist_c * post_t + \delta_2 * X_{ct} + \Upsilon_c + \lambda_t + \varepsilon_{ct} \quad (4)$$

Here, $Hypo_Route_Dist_c$ is simply the continuous distance from the centroid of the tract to the hypothetical route.

One potential criticism of the least-cost route is that the hypothetical route mapped out is not necessarily feasible without acquiring the rights-of-way. Metro must acquire the right-of-way to build a transit route on the land, and it is much easier to acquire these rights if they were previous streetcar or railroad lines in the past. For example, in 1990, Metro purchased 175 miles of rights-of-way for $450 million from Southern Pacific, including the 21-mile Burbank Branch line in San Fernando Valley (the future Orange line) and the 14-mile Exposition Boulevard route from downtown to Santa Monica (the future Expo line). Also, in 1990, the LACTC purchased the right-of-way for the Santa Fe railroad tracks to Pasadena (future Gold line). Thus, Metro builds on routes that have clearly defined and owned rights-of-way that are primarily based on the previous streetcar systems. The streetcar system appears to form the basis for most routes, with the exception of the Gold Line which uses the Santa Fe Railroad tracks and the extensions of some of the other routes, such as the Green line which uses a freeway median[26] for most

[25] When the treatment variable is binary, as is the case here, the instrument cannot be continuous or there will be a forbidden regression problem (Angrist and Pischke 2009)

[26] Interestingly, most of the Green line, before it veers south on the west end, is probably the best example of an exogenously determined route for the portion on Interstate 105. It was the result of a court settlement regarding acquisition of land for Century Freeway (Interstate 105) that required a stipulation requiring a median be built for

of the route and the last two stations in the southern portion of the Gold line that connects through East Los Angeles.[27] Therefore, I proxy for the universe of all rights-of-way that Metro could have conceivably acquired by mapping out the historical streetcar routes of Pacific Electric and the Los Angeles Railway as well as existing railroads in the United States (see Figure 6.1).[28]

One obvious potential instrument using the historical routes is to construct the hypothetical route by isolating the possible paths that connect the origin and destination areas with the historical rights-of-way. For example, to connect to Santa Monica in the West, it looks like there are multiple viable paths. Using historical routes as instruments is supported in the transportation literature to study transportation-related outcomes such as modern traffic patterns (Redding and Turner 2015; Duranton and Turner 2011). However, there are a few reasons why this is not the appropriate method to use in this context. First, and most importantly, IV is only appropriate if it meets the exclusion restriction. However, in this context, the distance of a tract to these early historical rights-of-way routes may affect contemporaneous labor market conditions, violating the exclusion restriction. It is conceivable that areas close to early historical railroads or the streetcar system led to economic development that persist over time and affect current labor market conditions. Brooks and Lutz (2019) demonstrate the persistence of the LA streetcar system on population densities, attributable to land use regulation, which lends credence to this idea of economic persistence. In addition to the exclusion restriction issue, it is not clear which historical routes should be chosen in constructing the hypothetical route that connects the origin and destination points – whether it should be the shortest connection, the one that services the most residents, or any possible route not longer than a certain threshold. Choosing too many potential routes leads to a weak first-stage. Therefore, I prefer to

transit on the highway. However, beyond the freeway, the Green line did not reach its goals. It failed to connect to the Norwalk Amtrak station due to opposition from Norwalk leaders, ran into limitations in connecting to LAX directly due to bureaucracy at the airport and ended up with a station 2 miles from LAX, and decided to go south to serve the Redondo Beach aerospace industry (which ultimately was severely crippled by the end of the Cold War when it was finally completed) (Elkin 2014)

[27] While the Orange line looks like it differs from the actual route placement, I attribute this to mapping error. The Orange line is supposed to be based on the Pacific Electric (Red Cars) route.

[28] It is assumed that all existing railroads are also based on historical rights-of-way. For example, the Gold line is based off the Santa Fe Railway built in 1859. These are primarily freight routes, and it can be assumed that Metro could have conceivable purchased the rights-of-way to convert to commuter lines.

use the least-cost route as the instrument since it yields a stronger first stage and satisfies the exclusion principle.

However, these historical routes are still relevant for determining the appropriate control tracts. Persistence of higher economic growth in neighborhoods that are close to these historical routes indicates that the IV estimates may be upward biased when using all tracts without Metro stations in Los Angeles County as the control group. Therefore, the control group is restricted to tracts that are close to the historical streetcar system or railroads, which I defined as a 0.8-kilometer buffer region (see Figure 6.B)..[29] I tested this theory using the public ACS data by determining whether there is a spurious effect by comparing the outcomes of tracts without Metro stations but part of the historical right-of-way against tracts without Metro stations and not part of a historical right-of-way and find that this is a reasonable restriction. The results of this analysis, as well as the overall analysis using public data, can be found in Appendix A.

Results:

Table 1 shows the TWFE/IV regression results matching equation (1). The TWFE estimate shows that living within walking distance of a Metro station is associated with a 0.45 percentage point increase in the tract-level proportion of residents who use rail or subway for commuting to work. The corresponding IV estimate indicates that the TWFE estimate is understated, and the IV estimates suggest a 1.02 percentage point tract-level increase in subway and rail usage. Thus, there is a clear uptick in usage of the Metro stations by residents resulting from these expansions. When we break out by the demographic groups (Black, Hispanic, and female), TWFE estimates range from 0.37 to 0.66. However, the IV estimates suggest that the commuting effect is weaker for the different demographic groups. For

[29] I only use the restricted group of control tracts in the binary specification, because I also make the 0.8-kilometer buffer region assumption in those specifications. It is not appropriate in the continuous specification of the treatment variable – although one might consider assigning different weights to tracts depending on their proximity to these historical routes.

Hispanic and female residents, the estimate is 0.74 and 0.92. The effect is not statistically significant for Black residents.

The effects of living within walking distance to Metro stations on the employment to population ratio for residents are strikingly clear. TWFE estimates suggest an increase of the employment to population rate by 0.94 percentage points for tracts with new stations within walking distance. There appears to be a larger employment effect for Blacks (2.44) and females (1.18). IV estimates suggest that the TWFE estimates are understated and suggest that there is an increase in employment to population by 5.96 percentage points. Similar to the TWFE estimates for Blacks and females, the IV employment estimates are higher than the general population (9.38 and 8.71 respectively). While the TWFE estimate for employment is not statistically significant for Hispanics, the IV estimate (4.26) is statistically significant at the 10% level.

For almost all demographic groups, there is no effect on population. This is likely partly due to our exclusion of those who migrated in the past year from the sample to reduce neighborhood sorting mechanisms. Additionally, there is an increase in the earnings of residents near Metro stations. The TWFE estimate suggests approximately a 3% increase in earnings for overall residents, a 17% increase in earnings for Black residents, and 4% increase in earnings for female residents. The IV estimates suggest that the earnings effect may be even higher, with IV estimates suggesting an approximate increase in 26% for overall residents, 23% increase in earnings for Hispanic residents, and 41% for female residents. Finally, for commute time, while TWFE are not significant, IV estimates imply that living within walking distance to a Metro station is associated with a 3.5-minute general decrease in commute time, with a larger decrease in commuting time for Hispanics by 4.19 minutes.

To summarize, Census tracts within walking distance of a Metro station are associated with higher residential employment. If we expect that the Metro stations are connecting residents to jobs that they otherwise would not be able to access, then we should expect an equivalent increase in subway and rail usage for work commutes. While there is a modest increase usage of subway or rail transit usage for

work commutes, the effect is not as large as the employment effect. One plausible explanation is that residents are not necessarily only joining the labor market due to the increased accessibility of the stations but that there is an increase in the development around the area which is connecting residents to more accessible jobs. This theory is mildly supported by shorter commute times for the average resident. Additionally, the increase in earnings indicates that residents are accessing higher paid jobs. Overall, these results suggest positive employment effects for residents.

Additionally, I test the effects of a continuous distance to a Metro station rather than a binary walking distance measure. This specification is interpreted as an increase in *proximity*, in other words a decrease in distance, from a Census tract to a Metro station to make the interpretation of the estimates easier. The TWFE estimates show that an 1% increase in proximity increases subway/rail usage by 1.25%, employment by 0.01%, and earnings by 0.306%. The IV estimates show that the TWFE are understated for employment and earnings, with estimates showing that a 1% increase in proximity is associated with a 1.91% increase in employment and 2.72% in earnings. However, the commute time is somewhat contrary to the binary specification; a 1% increase in proximity to a Metro station *increases* commuting time by 0.95%. This could be indicative that we have differential effects depending on the distances. Living within walking distance of Metro stations may be associated with increased development around these stations which allows residents to connect with more jobs near the stations and reduce their commuting time. However, as we get farther away from the Metro stations beyond the walking distance boundaries, residents tend to *increase* their commuting time from new Metro stations to either commute to the increased jobs near the Metro stations or use the Metro lines to access more distant jobs. Finally, we note the IV estimates show a 1% increase in proximity to Metro stations is associated with a 1% decrease in population.

When we break out the effects by different demographic groups, we observe the following patterns. For Blacks, from the IV estimate, we observe an increase of 2.1% employment to population ratio for every 1% increase in proximity to Metro stations, although this is only statistically significant at

the 10% level. For Hispanics, we find a smaller increase in the employment to population ratio by 1.5%. For females, we find the strongest effect on employment, with an 2.9% increase in employment. Overall, for all demographic groups, we find population decreasing as proximity to Metro stations increases, ranging from 1.1% for females to approximately 3% for Blacks and Hispanics. On earnings, we generally find positive income effects for Hispanics (1.9%) and females (3.6%) but negative income effects for Blacks (4.5%). We generally do not find any statistically significant results for commuting times, except for the IV regression for females, who see a 1% increase in commuting time for every 1% increase in proximity. We find statistically significant and positive IV estimates on subway and rail usage for Hispanics (1.8%) and females (1.3%) but not Blacks. However, the TWFE estimates for subway and rail usage are statistically significant across all demographic groups, ranging from 0.9% to 1.1%.

In general, IV estimates using the continuous specification are consistent with the binary specification. In both specifications, we generally see positive effects on employment to population ratios and increased subway and rail usage. This is generally indicative that Metro stations are associated with increased usage of these lines and increased employment. However, the employment effects being stronger than the subway and rail usage indicates that the increased employment may not necessarily be directly related to new workers using these stations to commute to work. Instead, increased development may be a strong contributing factor. In both specifications, we find positive earnings effects, particularly for Hispanics and females, but not for Blacks. However, there are some exceptions. The binary specification is associated with decreased commuting times while the continuous specification shows an increase in commuting time. Additionally, the continuous specification shows decreased population. However, recall that I do not apply the restriction of historical tracts in the continuous specification.

Overall, I have shown that there is robust evidence that Metro stations have a positive effect on residential employment. However, the lack of a strong uptick in usage of these lines suggests that the idea of new workers who were at the extensive margin becoming employed due to accessibility to distant jobs is not the case, or else the usage of these lines should increase at the same magnitude as employment.

Thus, a more likely story is if Metro stations lead to increased economic activity and development in neighborhoods near the stations, then the increased employment is a result of increased job density near the station.

The confidential ACS data also provides information about place of work. Thus, in theory, we can measure the effect of Metro stations on the job density by aggregating the number of workers in each tract per year. However, we note that the underlying survey sample sizes appear to be very small for each tract, because respondents either did not answer the place of work question or the data was suppressed for confidentiality concerns which is commonly done with the LODES data (Graham et al. 2014). We did not find any statistically significant results using the confidential ACS data on job densities (column 6).[30] I turn to the publicly available Longitudinal Employer-Household Dynamics Origin-Destination Employment Statistics (LODES) as a different measure for job density.

The results with the LODES data are more supportive of the theory that job density increased as a result of new Metro stations. Using the binary specification, the TWFE estimate shows that being within walking distance to a Metro station is associated with a 6% increase in jobs for the Census tract. The IV estimate shows that the TWFE estimate is understated and downward biased; the IV estimate indicates a 29% increase in jobs from being within walking distance to a Metro station. The binary specification does not yield statistically significant results for Hispanic or female workers in either the TWFE or the IV regression, but the IV estimate for Black workers shows an increased job density of 130%. We next turn to the continuous specification. Using a continuous measure of proximity rather than a binary walking distance measure may be more appropriate as it does not apply a walking distance threshold; we may expect increased commercial activity in the areas surrounding Metro stations that decreases as the distance from the Census tract to Metro stations decrease (or proximity increases) that goes beyond a 0.8-kilometer cutoff.

[30] We also measured (and were approved to release) the effect on earnings for jobs using the confidential ACS data but like job density, we did not find any meaningful results.

Using the continuous specification for proximity, we find increased job density in tracts, including across demographic groups. The TWFE estimates show that a 1% increase in proximity to a Metro station is associated with a 0.06% increase in overall jobs and a 0.09% increase in Black jobs. The IV estimate shows that these are understated; the increase in overall jobs is 0.36% and 0.55% for Black jobs. Additionally, although the TWFE estimates are not statistically significant for Hispanic or female jobs, they are positive and statistically significant in the IV regressions. A 1% increase in proximity to Metro stations increased Hispanic jobs by 0.58% and female jobs by 0.72%.

For the IV specification, we also report the Kleibergen-Paap rk Wald F statistic to measure the strength of the first stage. We note that the F-statistic differs because some tract-year observations are missing information for the dependent variable when Census suppresses output. For disclosure purposes and to avoid implicit samples, we were only allowed to release the rounded total tract-year observations for the entire LA County. The first stage F statistic is generally very strong, except for regressions using the number of jobs regression using the ACS in the binary specification (Table 1, col. 6). The continuous specification IV regressions also seem to have a much stronger first stage than the binary specification – with a F-stat of 199.8 for the full sample..[31] We also present the Overidentification test and endogeneity test, which are expressed in p-values. The endogeneity test generally shows that in most instances, the IV regression should be preferred over the TWFE.

To summarize, there is clear and convincing evidence that Metro stations lead to increased employment to population ratio for residents. There is also a positive earnings effect, for overall residents, Hispanics, and females. Thus, the construction of new Metro stations is associated with positive employment outcomes. Additionally, there is increased usage of subway and rail usage for work, indicating that newer Metro stations are being used to an extent for work-related commutes, but this uptick is not as large as the employment boost. Therefore, it is unlikely the employment boost can be

[31] Again, the first stage estimates are slightly different between each column due to different weights and control variables being used (i.e for the estimates on Black employment, I use Black population as weights and omit race controls).

solely explained by expanded usage of the new transit infrastructure. We find that increased residential employment is likely a result from increased job densities in areas near transit stations. Commuting times decrease for workers who live within walking distance to Metro stations, supporting this theory.

Conclusion:

This paper contributes to the literature on the effects of public transit infrastructure on employment outcomes and distinguishes itself by addressing route placement endogeneity using a least-cost instrument to predict route placement. The positive employment effects from proximity to Metro stations suggests that public transit can be an effective tool in reducing spatial mismatch by connecting residents to more jobs. This connection may be attributed to increased development in the area which attracts more jobs. As the nature of work shifts to more remote options in the post COVID-19 era, investment in public transit infrastructure can still be useful by having a positive impact on development and job growth.

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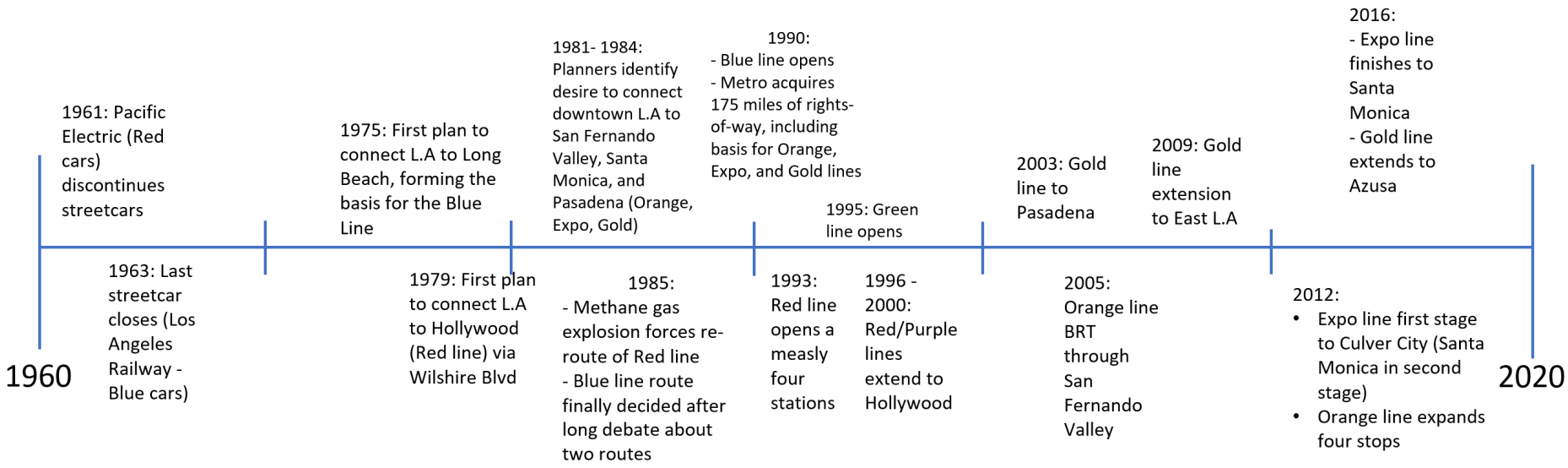
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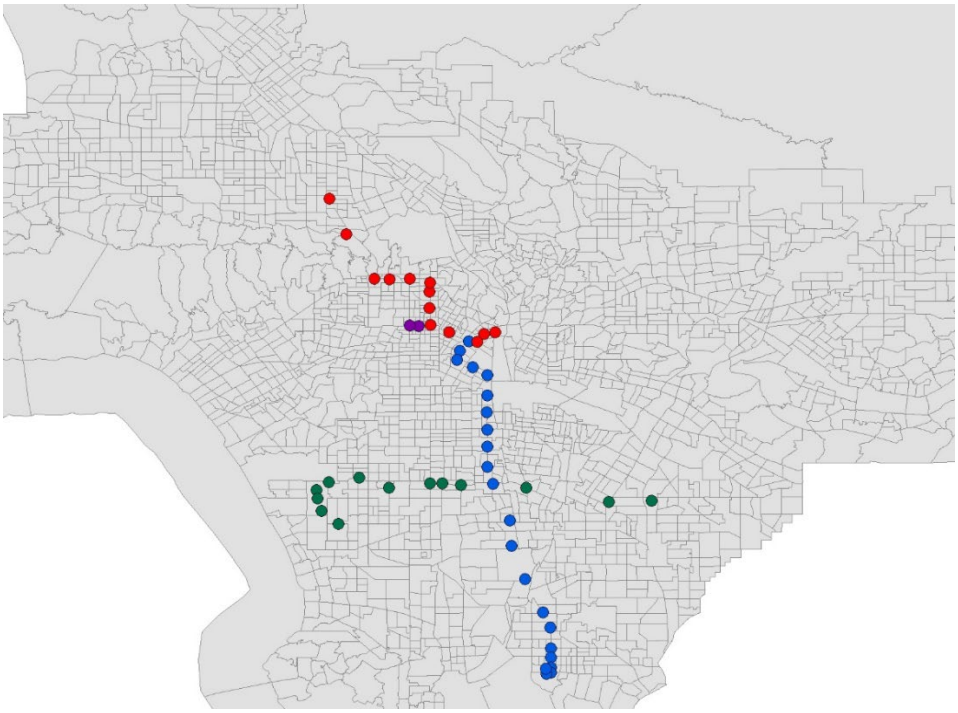
Figure 1: Timeline of the Los Angeles Metro:



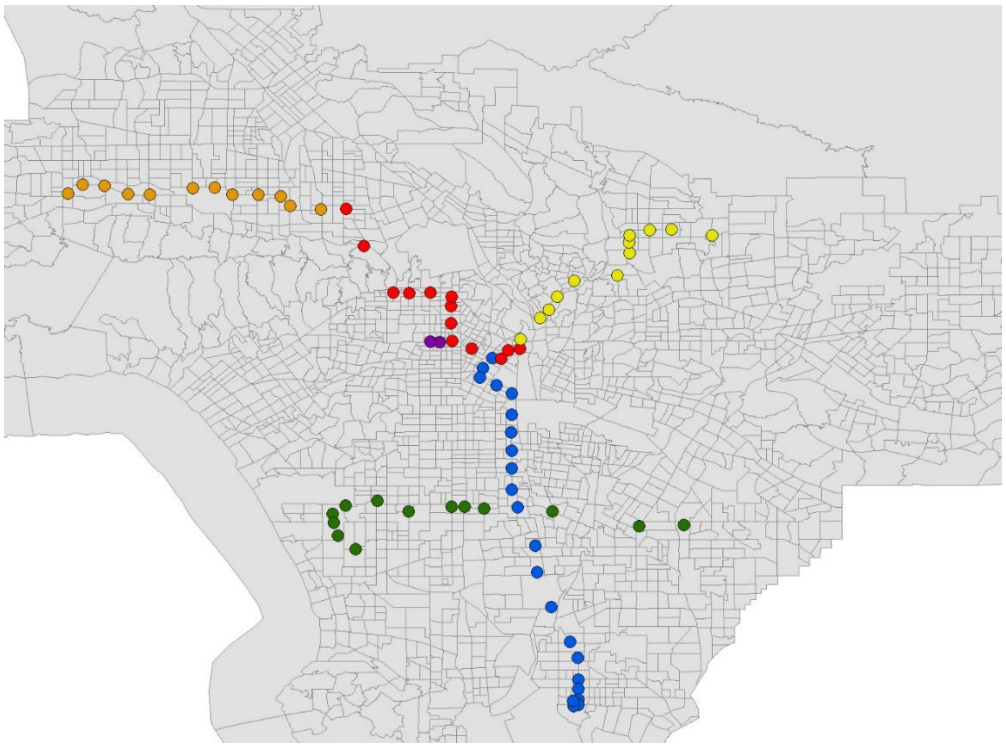
Sources: Elkin (2014) and L.A. Metro

Figure 2: Los Angeles Metro Stations in 2000, 2005, 2012, 2016, and planned 2026 routes

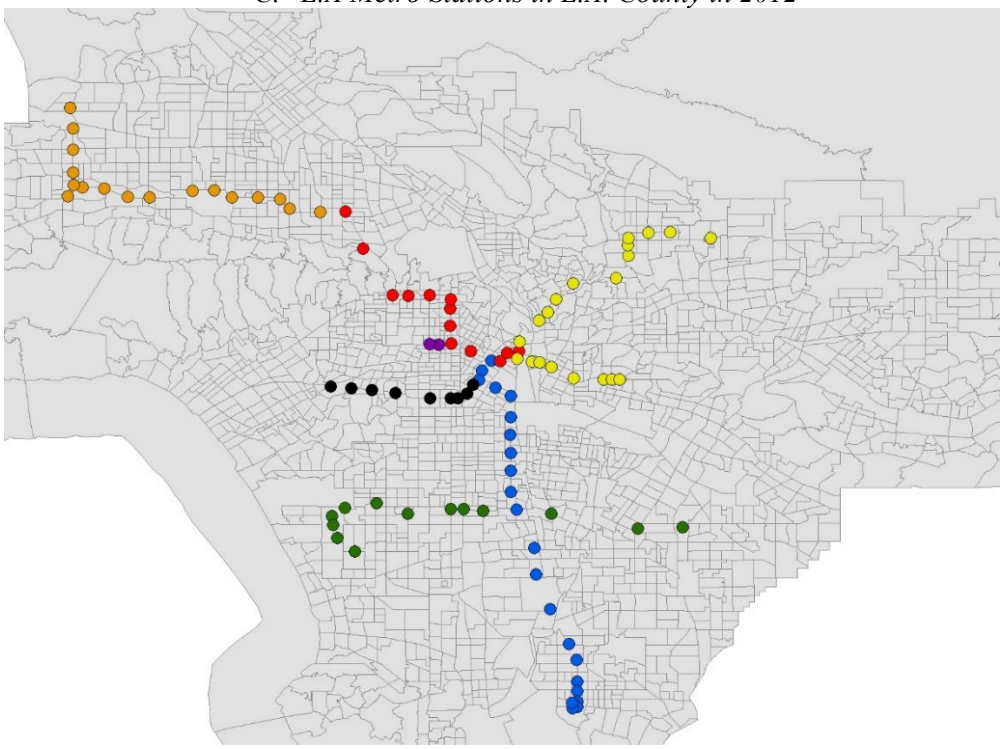
A. L.A Metro Stations in L.A. County in 2000



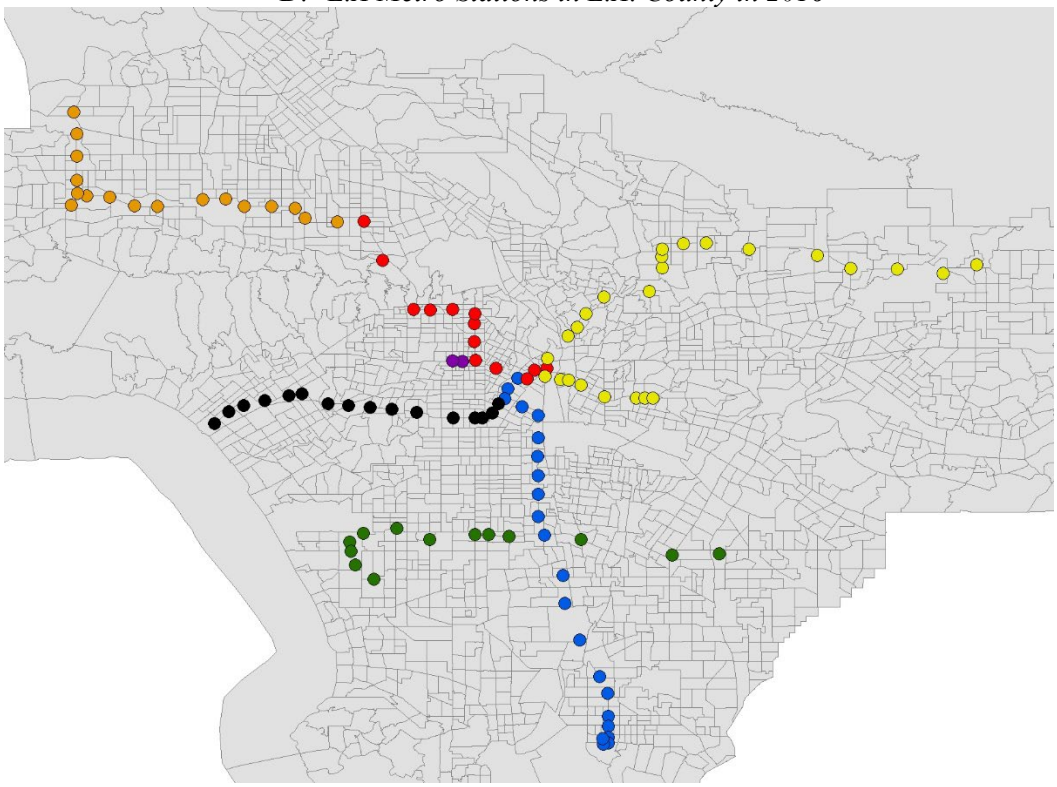
B. L.A Metro Stations in L.A. County in 2005



C. L.A Metro Stations in L.A. County in 2012

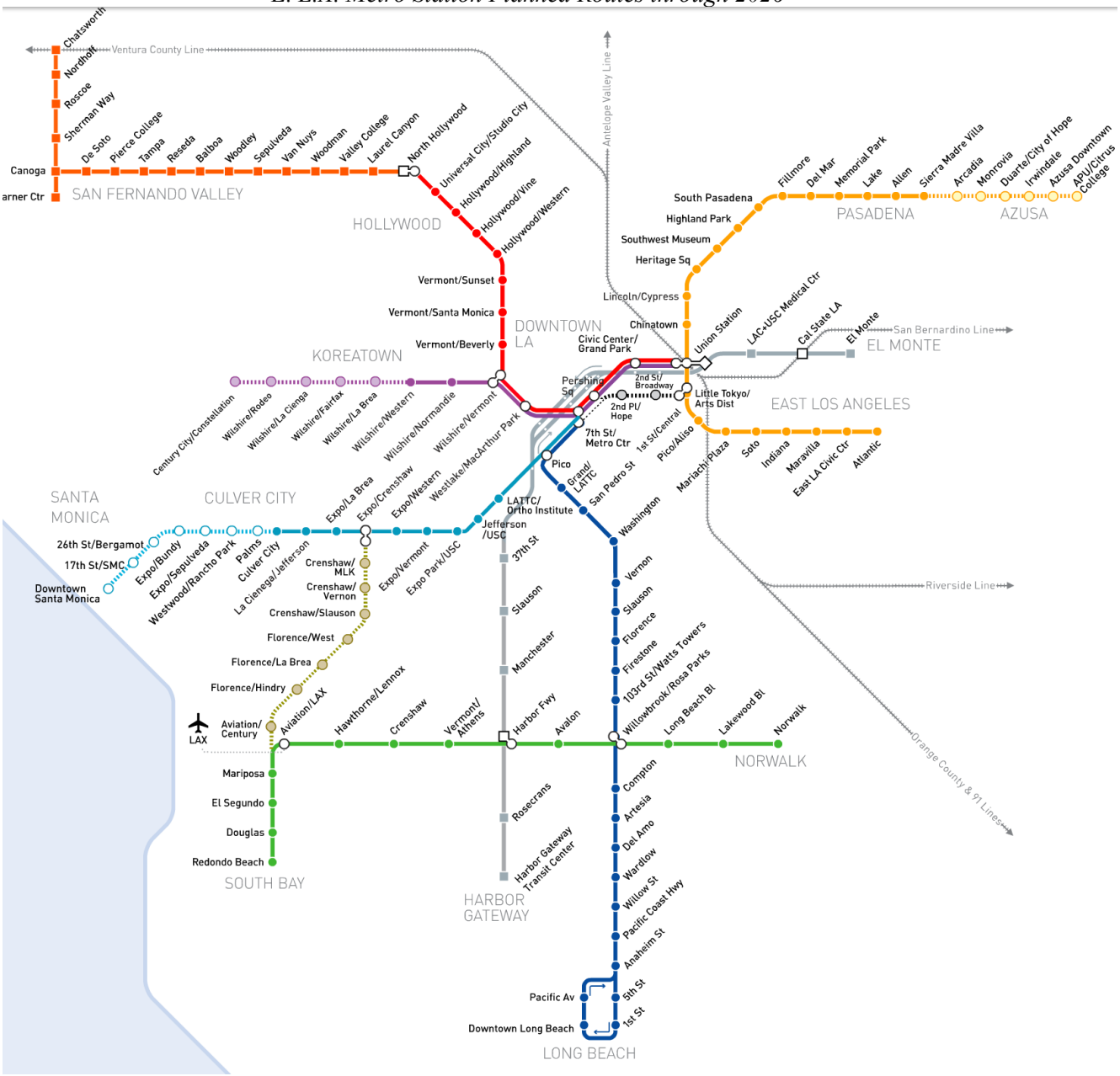


D. L.A Metro Stations in L.A. County in 2016



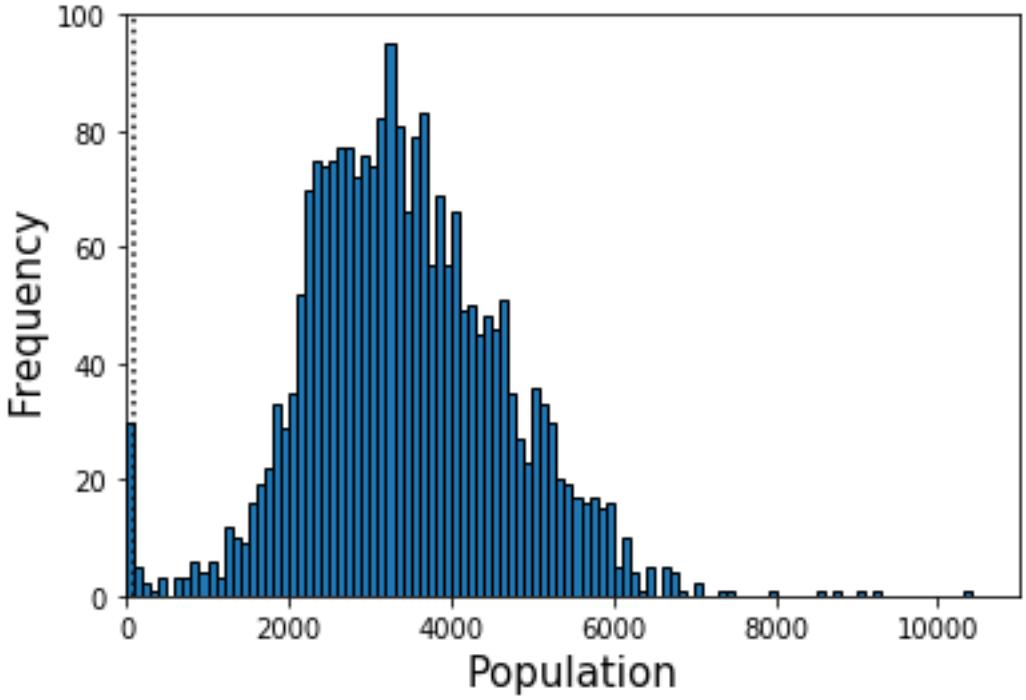
Source: Los Angeles Metropolitan Transit Authority overlaid on Los Angeles County shape file. The black routes are the Expo line and the yellow is the Gold line – all other routes are colored according to their name.

E. L.A. Metro Station Planned Routes through 2026



Source: Los Angeles Metropolitan Transit Authority.

Figure 3: Total Tract Population (16+) from 2012-2016



Source: American Community Survey (2012-2016). Bin size is set at 100.

Figure 4: Early Planned Routes

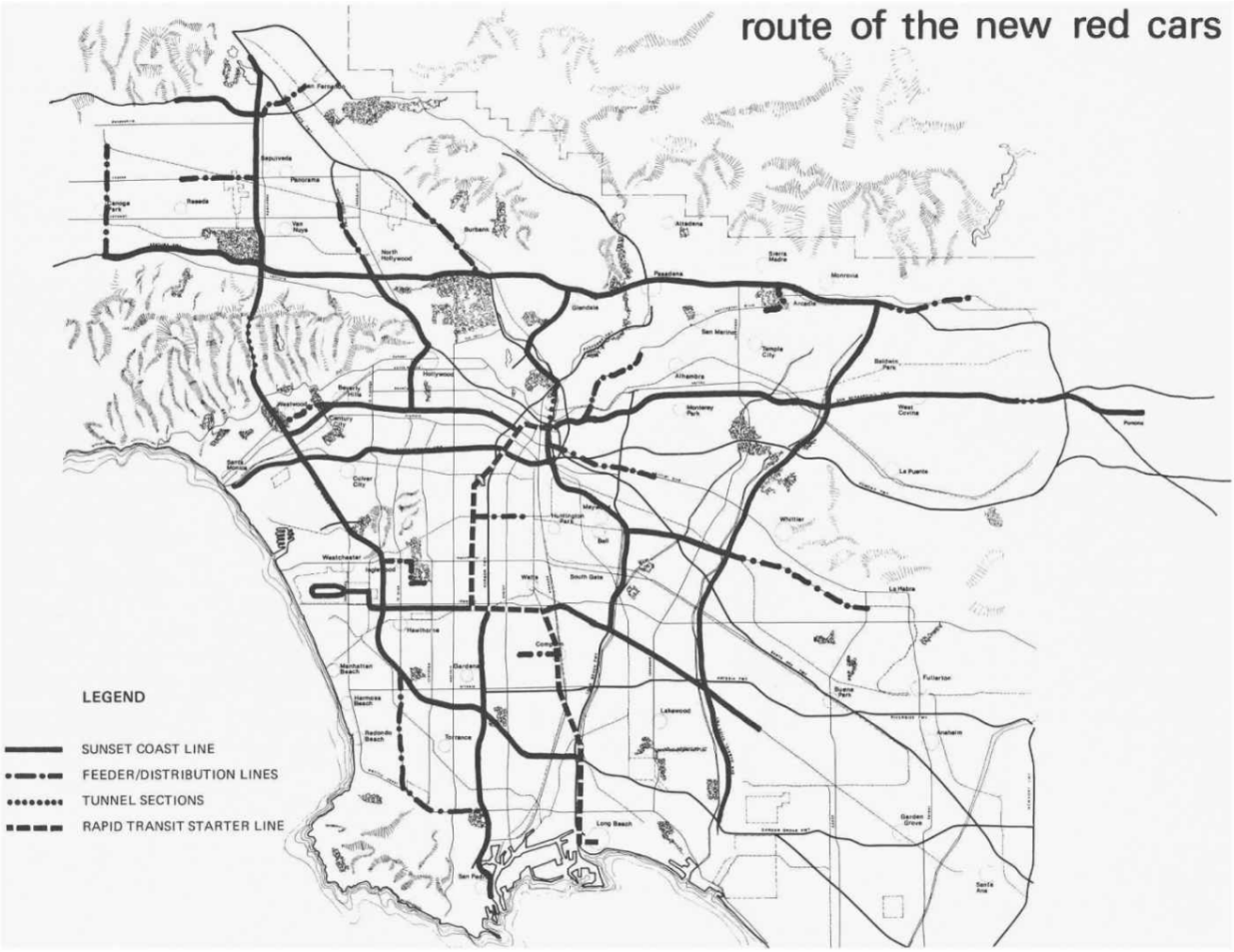


FIGURE 5. Los Angeles county supervisor Baxter Ward's proposed Sunset Coast Rapid Transit Line in 1976. (Courtesy of the Los Angeles County Metropolitan Transportation Authority Research Library and Archive. Reprinted with permission.)

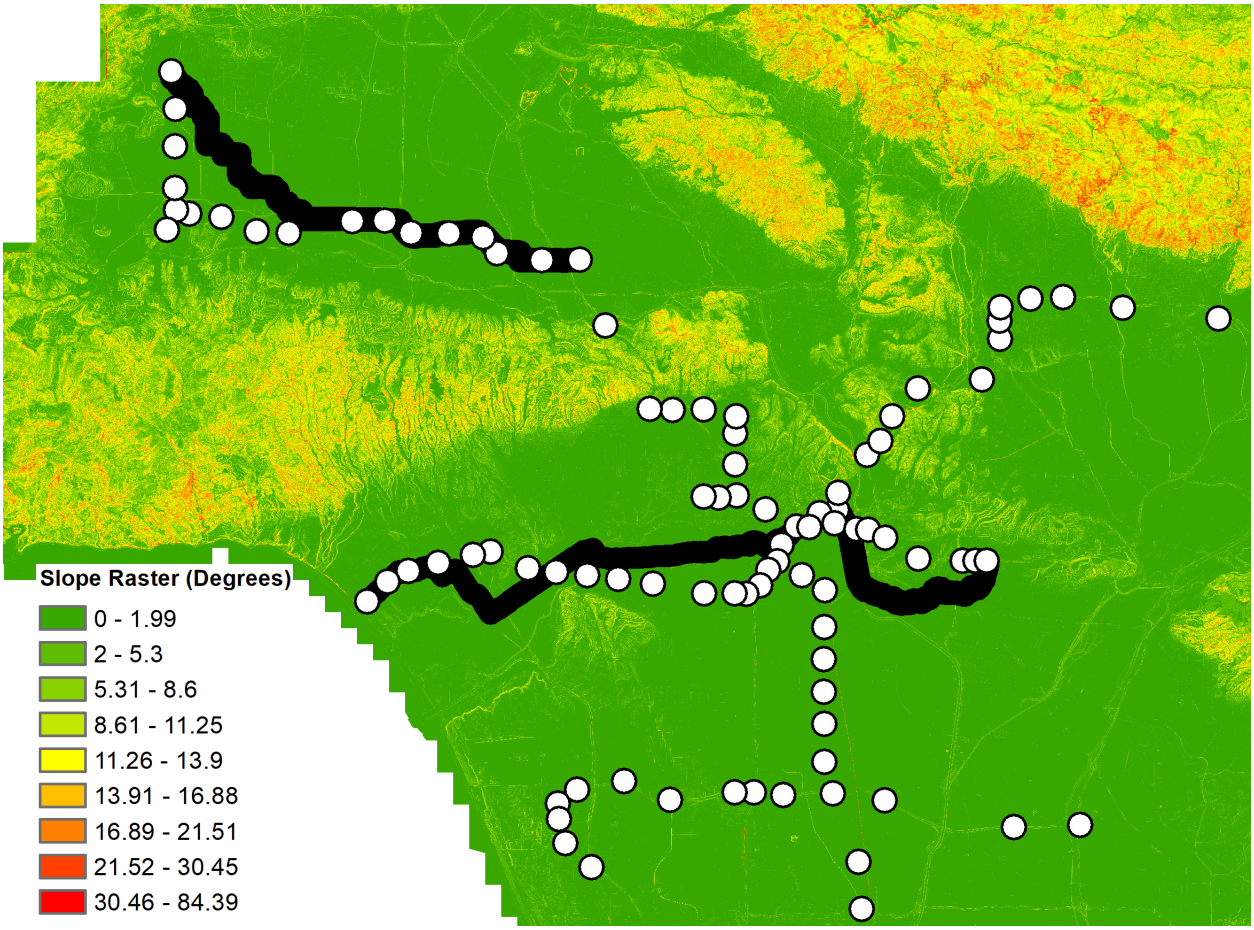


Source: Ballot Proposition A. November 4, 1980

Figure 1-2 Regional Rail Rapid Transit System

Source: Copied from Elkin (2014)

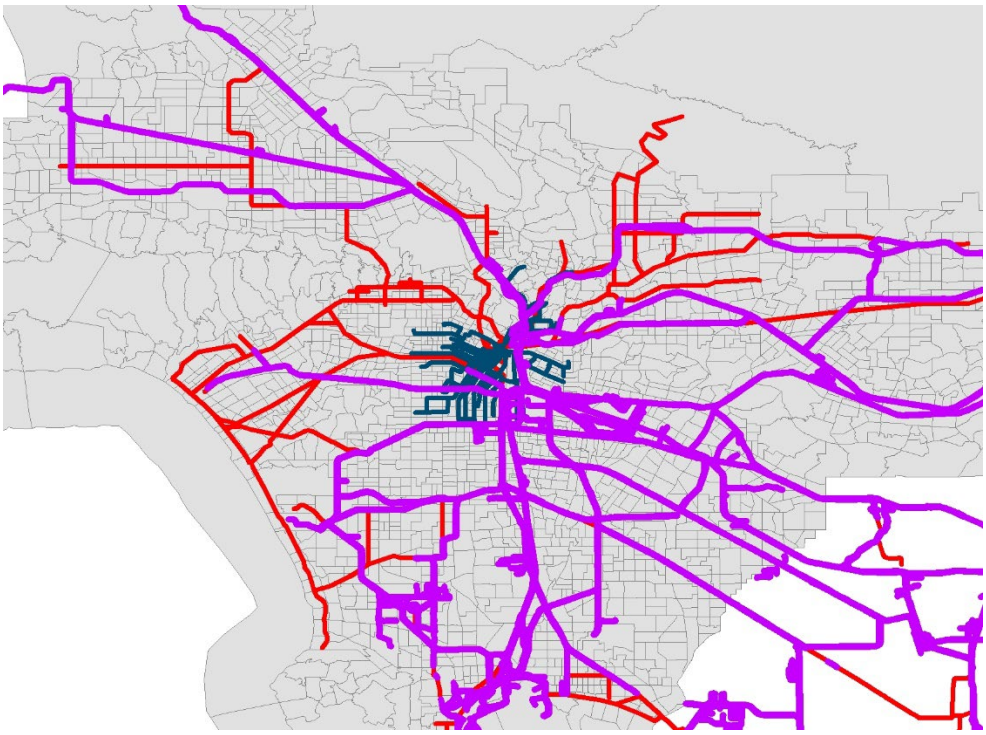
Figure 5: Hypothetical Least-Cost Route for Origin-Destination



Source: Los Angeles County GIS Data Portal. The digital elevation model is converted to a slope raster (shown in the background) and the least-cost route is calculated using spatial analysis tools in ArcMap that account for elevational slope changes.

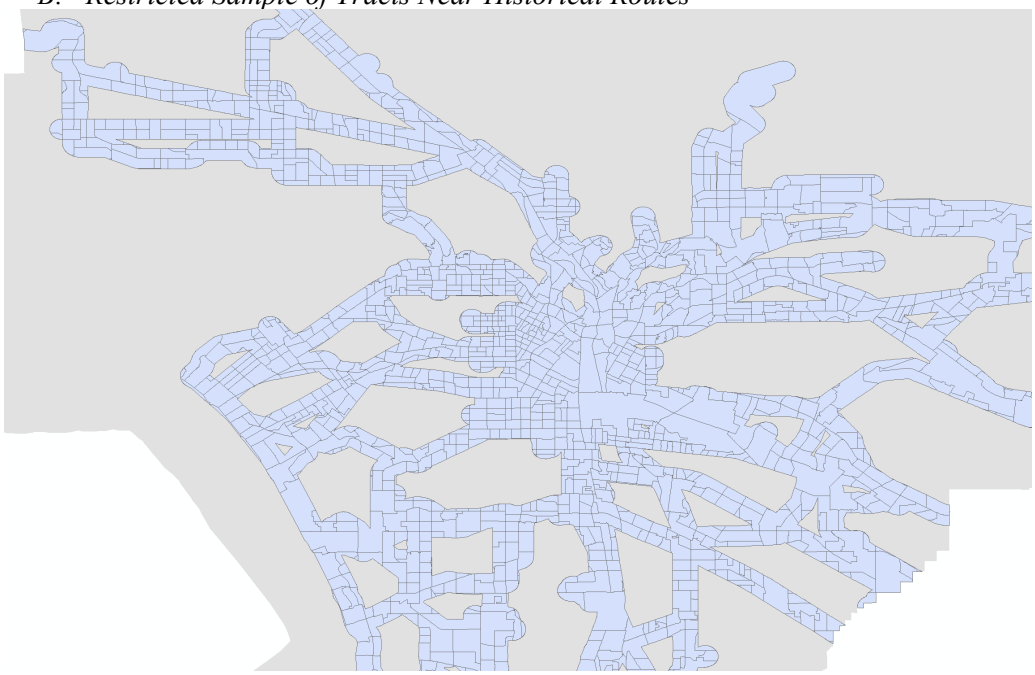
Figure 6: Rights-of-Way Based on Historical Routes

A. Red Cars, Yellow Cars, and Currently Operating Railroads



The red routes indicate a Pacific Electric streetcar route (Red cars). The blue routes indicate a Los Angeles Railway route (Yellow cars). The purple indicates a currently operating railroad.

B. Restricted Sample of Tracts Near Historical Routes



This shows a map of tracts that are counted in the restricted subsample which is limited to a 0.8-kilometer walking distance around the routes above.

Table 1: The Effect of Metro Stations (Binary Specification)

Variables	(1) Emp	(2) Pop	(3) Sub/Rail Usage	(4) Earnings	(5) Commute Time	(6) Number of Jobs (ACS)	(7) Number of Jobs (LODES)
I. All							
TWFE: Metro	0.94**	-0.00	0.45***	0.03**	-0.54	0.03	0.06*
	[0.48]	[0.01]	[0.07]	[0.01]	[0.33]	[0.04]	[0.03]
IV: Metro	5.96***	0.06	1.02***	0.26***	-3.50**	0.15	0.29**
	[1.99]	[0.06]	[0.26]	[0.05972]	[1.40]	[0.30]	[0.14]
First-Stage F-Stat	32.80	32.80	32.80	32.80	32.80	10.90	32.80
Overidentification Test	0.714	0.080	0.751	0.838	0.226	0.918	0.266
Endogeneity Test	0.008	0.169	0.018	0.000	0.019	0.543	0.150
II. Black							
TWFE: Metro	2.44*	0.01	0.66***	0.17**	1.06	0.01	0.09
	[1.43]	[0.04]	[0.22]	[0.07]	[1.05]	[0.07]	[0.07]
IV: Metro	9.38***	0.03	0.19	0.20	1.89	0.23	1.30***
	[3.44]	[0.10]	[0.51]	[0.23]	[2.49]	[0.36]	[0.49]
First-Stage F-Stat	20.47	20.47	20.47	20.47	20.47	11.63	32.80
Overidentification Test	0.221	0.547	0.122	0.537	0.362	0.697	0.080
Endogeneity Test	0.029	0.744	0.766	0.861	0.457	0.376	0.018
III. Hispanic							
TWFE: Metro	0.32	-0.00	0.37***	-0.00	-0.25	0.02	-0.01
	[0.75]	[0.02]	[0.08]	[0.02]	[0.43]	[0.05]	[0.07]
IV: Metro	4.26*	-0.13*	0.74***	0.23***	-4.19**	0.27	0.75
	[2.35]	[0.08]	[0.28]	[0.07]	[1.77]	[0.32]	[0.46]
First-Stage F-Stat	23.79	23.79	23.79	23.79	23.78	9.38	32.80
Overidentification Test	0.477	0.214	0.426	0.535	0.018	0.766	0.214
Endogeneity Test	0.083	0.133	0.165	0.000	0.011	0.239	0.141
IV. Female							
TWFE: Metro	1.18*	-0.03*	0.49***	0.04**	-0.19	0.03	0.04
	[0.67]	[0.02]	[0.09]	[0.02]	[0.44]	[0.04]	[0.05]
IV: Metro	8.71***	0.03	0.92***	0.41***	-1.41	0.09	0.56
	[2.61]	[0.07]	[0.2798]	[0.10]	[1.79]	[0.32]	[0.40]
First-Stage F-Stat	33.21	33.21	33.21	33.21	33.20	11.30	32.80
Overidentification Test	0.505	0.108	0.894	0.460	0.344	0.716	0.241
Endogeneity Test	0.002	0.251	0.130	0.000	0.475	0.591	0.274

N (the number of tract-year observations) is approximately 25,000. Metro is a binary variable that takes the value of 1 if the tract-year contains a Metro station within walking distance, 0 otherwise. Emp (col 1) is expressed as the employment to population ratio. Sub/Rail Usage (col 3) is expressed as the percentage of the population who use subway or rail to commute to work. Commute time (col 5) is expressed in minutes. Population (col 2), earnings in dollars (col 4), and number of jobs (Col 6 and 7) is transformed into the inverse arcsine. The First-Stage F statistic is the Kleibergen-Paap rk Wald F statistic. The F-statistic differs because some tract-year observations are missing information for the dependent variable when Census suppresses output. For disclosure purposes and to avoid implicit samples, we were only allowed to release the overall tract-year count for the entire LA County. The Overidentification test and endogeneity test are expressed in p-values. Standard errors are clustered by tract. *, **, and *** denote 0.1, 0.05, and 0.01 significance levels. This table was approved for release by the U.S. Census Bureau under CBDRB-FY23-P2146-R10234.

Table 2: The Effect of Metro Stations (Continuous Specification)

Variables	(1) Emp	(2) Pop	(3) Sub/Rail Usage	(4) Earnings	(5) Commute Time	(6) Number of Jobs	(7) Number of Jobs (LODES)
I. All							
TWFE: Proximity	0.010*	-0.062	1.253***	0.306***	-0.058	0.443*	0.058**
	[0.052]	[0.080]	[0.182]	[0.074]	[0.064]	[0.267]	[0.023]
IV: Proximity	1.910***	-0.996**	0.285***	2.715***	0.951***	0.440	0.363***
	[0.270]	[0.423]	[0.077]	[0.417]	[0.304]	[1.386]	[0.103]
First-Stage F-Stat	199.8	199.8	199.8	199.8	199.8	75.08	197.7
Endogeneity Test	0.000	0.024	0.035	0.000	0.001	0.998	0.002
II. Black							
TWFE: Proximity	0.145	-0.010	1.065***	-0.473	-0.012	0.075	0.086*
	[0.245]	[0.222]	[0.308]	[0.433]	[0.165]	[0.516]	[0.048]
IV: Proximity	2.097*	-3.005***	1.180	-4.473**	-0.012	-1.217	0.553**
	[1.163]	[0.986]	[0.983]	[1.903]	[0.773]	[1.960]	[0.275]
First-Stage F-Stat	86.25	86.25	86.25	86.25	86.23	57.96	197.7
Endogeneity Test	0.083	0.001	0.904	0.031	1.000	0.504	0.074
III. Hispanic							
TWFE: Proximity	0.100	-0.129	0.921***	0.207*	-0.041	0.317	-0.039
	[0.081]	[0.122]	[0.167]	[0.115]	[0.092]	[0.253]	[0.049]
IV: Proximity	1.473***	-3.126***	1.780***	1.870***	0.394	-0.574	0.576**
	[0.385]	[0.615]	[0.664]	[0.512]	[0.387]	[1.290]	[0.271]
First-Stage F-Stat	134.2	134.2	134.2	134.2	134.2	75.10	197.7
Endogeneity Test	0.000	0.000	0.187	0.001	0.242	0.490	0.017
IV. Female							
TWFE: Proximity	0.080	-0.176*	0.911***	0.230**	-0.067	0.573*	0.038
	[0.090]	[0.096]	[0.159]	[0.117]	[0.082]	[0.305]	[0.037]
IV: Proximity	2.928***	-1.105**	1.284*	3.600***	0.952**	0.559	0.719***
	[0.469]	[0.450]	[0.713]	[0.667]	[0.419]	[1.377]	[0.249]
First-Stage F-Stat	198.8	198.8	198.8	198.8	198.8	76.17	197.7
Endogeneity Test	0.000	0.035	0.594	0.000	0.013	0.992	0.004

N (the number of tract-year observations) is approximately 32,500. Proximity is the negative distance in kilometers from the centroid of the tract to the nearest Metro station, expressed as an inverse arcsine transformation. All dependent variables (col 1-6) are transformed using the inverse arcsine. The First-Stage F statistic is the Kleibergen-Paap rk Wald F statistic. The F-statistic differs because some tract-year observations are missing information for the dependent variable when Census suppresses output. For disclosure purposes and to avoid implicit samples, we were only allowed to release the overall tract-year count for the entire LA County. The endogeneity test is expressed in p-values. Standard errors are clustered by tract. *, **, and *** denote 0.1, 0.05, and 0.01 significance levels. This table was approved for release by the U.S. Census Bureau under CBDRB-FY23-P2146-R10234.

Appendix A: Analysis Using the 5-year ACS Public Data

An earlier iteration of this paper used publicly available data from the American Community Survey (ACS) prior to updating to confidential ACS data. The public data contains averages for labor market outcomes within a tract, such as labor force participation, employment to population ratio, median income earned in the past 12 months, and number of workers who use public transit, broken down by bus and subway/rail. Additionally, I retrieve demographic data such as race, gender, education, migration, and vehicle ownership as controls. The main limiting factor with the publicly available ACS data is that it contains information at the census tract level for aggregated 5-year samples only, for confidentiality concerns. For example, the employment data reported in 2009 contain the average employment by tract from 2005-2009. This is limiting in that it only allows for one pre-intervention and one post-intervention period. Therefore, I evaluate the 2009 and 2012 expansions by using the 2005-2009 as the pre-period and 2012-2016 as the post period.[32] Along with not being able to evaluate the effects of the 2016 Expo line expansion, the public data also did not allow us to drop individuals who migrated in the past year.

I modified the TWFE model to a generalized difference-in-differences model that considers the effects from two types of stations in two periods – existing stations in operation in the pre-period and new stations that began operation in the post period.[33] The equation is as follows:

$$Y_{ct} = \alpha + \beta_1 * New_c * post_t + \beta_2 * Existing_c * post_t + \beta_3 * X_{ct} + post_t + \Upsilon_c + \varepsilon_{ct} \quad (2)$$

In this specification, New_c indexes whether a tract c contains or will contain a new Metro station within walking distance which is multiplied by $post_t$, an indicator for the post-treatment period after the station is built. $Post_0$ is defined as the time period from 2005-2009 and $Post_1$ is defined as the time period from 2012-2016 in order to evaluate a total of three expansions in 2009 and 2012 - the Gold line light rail

[32] I also tested evaluating each Metro expansion in 2009 and 2012 separately (i.e. using 2010-2014 as the post period for the 2009 expansion and 2007-2011 as the pre-period for the 2012 expansions) but along with issues regarding overlapping Metro expansions, for the most part, the results were not too different.
[33] In two periods, the $post_t$ interaction fully accounts for the increase in the size of the network.

expansion from Union Station to East Los Angeles, an expansion to the Orange line BRT, and the first leg of the Expo light rail connecting downtown to Culver City. $Existing_c$ indexes whether tract c contains an existing Metro station. Including $Existing_c$ accomplishes two goals. First, it accounts for the network effect in tracts with existing stations in the pre-period since there is a treatment effect of expanding the network in these tracts due to the new Metro stations that increase accessibility across the Metro system. Additionally, this ensures that the appropriate control group is restricted to tracts that do not contain any existing Metro stations. Therefore, two effects are being identified in equation (2); β_1 identifies the effect of the Metro expansion in tracts that receive a new Metro station relative to tracts that do not have any stations, and β_2 measures the effect of the Metro expansion in tracts that already contain existing Metro stations also relative to tracts that do not have any stations. If a tract is within walking distance of a new Metro station and an existing Metro station, I consider it as *Existing* rather than *New*. The intuition is that although there is a new station in this tract, residents were already connected to the system and the benefit to them is the increased areas for access rather than gaining access to the system, as a new station would provide. Finally, I eliminate tracts that are within walking distance of Metro stations built in the 2016 expansion from the control group.

In the binary specification, I restrict the control tracts to tracts that were part of historical the historical streetcar system or railroads, which I defined as a 0.8-kilometer buffer region. Persistence of higher economic growth in neighborhoods that are close to these historical routes indicates that the IV estimates may be upward biased when using all tracts without Metro stations in Los Angeles County as the control group. I test this theory by determining whether there is a spurious effect by comparing the outcomes of tracts without Metro stations but part of the historical right-of-way against tracts without Metro stations and not part of a historical right-of-way with the following equation:

$$Y_{ct} = \alpha + \beta_1 * Historical_c * post_t + \beta_2 * X_{ct} + post_t + \Upsilon_c + \varepsilon_{ct} \quad (6)$$

In this specification, $Historical_c$ indicates whether a tract is within a 0.8-kilometer walking distance buffer around the historical streetcar route or railroad. All tracts that will contain Metro stations within walking distance are omitted. If the theory is true, then β_1 should generate a positive, spurious effect.

I also employ a continuous distance-based treatment variable that measures the effect of proximity to a Metro transit station on employment in equation (3).

$$Y_{ct} = \alpha + \beta_1 * Proximity_{ct} + \beta_2 * X_{ct} + \Upsilon_c + post_t + \varepsilon_{ct} \quad (3)$$

As with the main analysis, $Proximity_{ct}$ indexes the negative distance in kilometers from the centroid of the tract *c* to the nearest Metro station in time *t*.

The results can be found in Appendix Table A1 and A2. For the most part, our analysis on the public data led to similar results as with the confidential data. The results indicated that increased proximity to Metro stations is associated with increased employment and higher usage of subway and rail. However, we note that we had evidence of lower earnings with the binary specification.[34]

[34] We also conducted the LODES analysis for number of jobs using the two-period specification. These results were generally inconclusive. These results are available upon request.

Appendix Table A1: Effect of Metro Stations on Labor Force Participation, Commuting, and Earnings (Binary)

Variables	(1) Subway or Rail	(2) Bus	(3) LFP (All)	(4) LFP (Black)	(5) LFP (Hisp)	(6) LFP (< HS)	(7) Emp (All)	(8) Emp (Black)	(9) Emp (Hisp)	(10) Emp (< HS)	(11) Earnings
I. DiD - Historical Route Test											
Historical*post	-0.01	-0.05	1.24***	2.77**	0.52	1.63**	0.97***	2.60**	0.20	1.81**	361.12
	[0.03]	[0.09]	[0.29]	[1.08]	[0.48]	[0.76]	[0.28]	[1.07]	[0.51]	[0.77]	[390.30]
Observations	3600	3600	3600	3094	3596	3538	3600	3094	3354	3538	3596
II. DiD - Main Specification											
New*post	0.36***	-0.66**	1.42**	3.48**	1.74**	-0.32	1.52***	3.86**	2.00**	0.07	-1164.38*
	[0.08]	[0.29]	[0.59]	[1.71]	[0.81]	[1.31]	[0.58]	[1.64]	[0.90]	[1.39]	[597.68]
Existing*post	0.40***	-0.11	1.64***	2.23*	1.71***	1.47**	1.76***	2.57**	2.03***	1.95***	143.64
	[0.06]	[0.19]	[0.36]	[1.19]	[0.48]	[0.73]	[0.37]	[1.20]	[0.55]	[0.75]	[355.57]
	[0.30]	[1.22]	[2.67]	[7.67]	[3.18]	[3.81]	[2.62]	[7.92]	[3.49]	[3.98]	[2502.15]
III. IV - Main Specification											
New*post	0.48**	-0.94	5.49***	4.51	4.88***	-0.00	4.93***	7.32**	4.31**	-0.90	-3634.59**
	[0.21]	[0.83]	[1.63]	[3.07]	[1.72]	[2.75]	[1.60]	[3.11]	[1.95]	[2.92]	[1634.36]
Existing*post	0.41***	-0.13	1.87***	2.33*	1.93***	1.50**	1.95***	2.92**	2.18***	1.87**	5.78
	[0.06]	[0.19]	[0.37]	[1.22]	[0.49]	[0.73]	[0.38]	[1.22]	[0.56]	[0.75]	[366.85]
First-Stage F-Stat	32.54	32.54	32.54	19.42	30.66	29.53	32.54	19.42	26.36	29.53	32.53
Endogeneity Test (p-value)	0.448	0.626	0.003	0.544	0.027	0.978	0.009	0.200	0.131	0.748	0.112
Overidentification Test (p-value)	0.129	0.074	0.639	0.490	0.580	0.185	0.263	0.177	0.380	0.349	0.843
Observations	3458	3458	3458	3016	3456	3416	3458	3016	3160	3416	3452

Source: American Community Survey (ACS) 5-year summary files, 2005-2009 ACS and 2012-2016 summary ACS files. This table reports DiD/IV estimates of equation (2) using the 0.8-kilometer walking distance buffer around Metro stations. *Historical* indexes whether a tract is within walking distance (0.8-kilometers) of a historical route or current railroad route. Tracts that had expansions in 2009 and 2012 are considered *New* while tracts that have existing stations from 1990-2005 are considered *Existing*. Tracts with expansions in 2016 are omitted from the regression. In column (1), (2), (6), and (10), the specification includes controls for shares female, Black, Asian, two or more races, Hispanic, high school degree, some college, bachelor's degree or higher, out of county migration in the past year, out of state migration in the past year, no vehicles, and one vehicle. Column (3), (4), (7), and (8) contain the same specification as (1) but exclude shares Black, Asian, two or more races, and Hispanic. Columns (5) and (9) contains the same specifications as (1) except without shares with high school degree, some college, and bachelor's degree or higher. All columns include tract fixed effects and a post period dummy variable. Regression is weighted by their respective population (i.e. all population 16 and over for LFP (All), etc. The first stage F-Stat differs due to the sample size data arising from censored data in omitted tracts for subgroups. The units for columns (1)-(10) represent percentage points from 0-100, and the units for column (11) is in US dollars. Standard errors are clustered by tract. *, **, and *** denote 0.1, 0.05, and 0.01 significance levels.

Appendix Table A2: Effect of Proximity to Metro Stations on Labor Force Participation, Commuting, and Earnings

Variables	(1) Subway or Rail	(2) Bus	(3) LFP (All)	(4) LFP (Black)	(5) LFP (Hisp)	(6) LFP (< HS)	(7) Emp (All)	(8) Emp (Black)	(9) Emp (Hisp)	(10) Emp (< HS)	(11) Earnings
1. DiD											
KmFromStation	0.136***	-0.075**	0.009*	-0.005	0.014**	0.004	0.009	0.003	0.016*	-0.003	0.004
	[0.029]	[0.030]	[0.005]	[0.019]	[0.007]	[0.011]	[0.006]	[0.020]	[0.008]	[0.014]	[0.009]
II. IV											
KmFromStation	0.401***	0.172	0.199***	0.273***	0.190***	0.148***	0.186***	0.246***	0.171***	0.156***	-0.021
	[0.091]	[0.121]	[0.025]	[0.064]	[0.032]	[0.044]	[0.026]	[0.067]	[0.040]	[0.051]	[0.031]
First-Stage F Stat	203.76	203.76	203.76	87.25	136.76	123.09	203.76	87.25	129.43	123.09	203.38
Endogeneity Test (p-value)	0.003	0.031	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.001	0.395
Observations	4604	4604	4604	4028	4600	4532	4604	4028	4248	4532	4598

Source: American Community Survey (ACS) 5-year summary files, 2005-2009 ACS and 2012-2016 summary ACS files. This table reports OLS/IV estimates of equation (3) using a continuous treatment variable, *KmFromStation*, which measures how far the centroid of the tract is from the nearest Metro station, expressed in its inverse arcsine form. The outcome variables in columns (1)-(10) are also expressed in inverse arcsines. See Table 1 for description of control variables. All columns include tract fixed effects and a post period dummy variable. Standard errors are clustered by tract. The pre-period defined here is 2005-2009 and the post period is 2012-2016.The first stage estimates differ due to sample size differences from censored data in omitted tracts for subgroups. *, **, and *** denote 0.1, 0.05, and 0.01 significance levels.

Appendix B: LODES Data on the Overall "Connectedness" of the Route

The origin-destination pairs in the LODES data can be used to evaluate the effect of the overall connectedness of the Metro system in residency to workplace pairs. To do this, I redefine the panel variable from tracts to all origin-destination tract-pairs of every worker that either lives or works in L.A. County.[35] Then, I can evaluate the effect of a connected route, defined as when the tract of residency and the tract of work both contain a Metro station, on the number of connected jobs. I use the entirely of the LODES data from 2002-2016 to evaluate multiple expansion routes. A proposed specification for this is as follows:

$$Y_{zt} = \alpha + \beta_1 * \text{WorkMetro}_{zt} * \text{LiveMetro}_{zt} + \beta_2 \text{LiveMetro}_{zt} + \beta_3 \text{WorkMetro}_{zt} + \Upsilon_z + \lambda_t + \varepsilon_{ct} \quad (1)$$

LiveMetro_{zt} indexes whether the origin, or place of residency, of an origin-destination pair *z* contains a Metro station within walking distance in time *t* and WorkMetro_{zt} indexes whether the destination, or place of work, of the origin-destination pair *z* contains a Metro station within walking distance in time *t*. When both LiveMetro_{zt} and WorkMetro_{zt} are equal to one, the origin-destination pair contains a Metro station in both the place of residency and work. Therefore, β_1 identifies the overall effect of having a connected route from residency to work on the number and type of jobs. Υ_z are dummy variables for all origin-destination pairs and λ_t are year dummy variables from 2002-2016.

There are 139,588,875 origin-destination commuting pairs rather than by tract to connect both commuting characteristics and the number and type of job (Appendix Table A). This table shows that the connectedness of the route matters. When origin-destination pairs contains a Metro station in the place of residence and work, on average, the total number of commutes along this route increases by 0.023. However, when the route is not connected but contains either a place of residency with a Metro station or

[35] The connection between Tract 1 and Tract 2 shows up as two origin-destination pairs due to the commuting occurring in two directions.

a place of work with a Metro station, there are generally statistically significant decreases in the number of commutes. Thus, these results show that the connectedness of the route matters.

Appendix Table B1: Metro on Number and Type of Jobs (for all Origin-Destinations)

Variables	Total Jobs	Low Inc. Jobs	Med Inc. Jobs	High Inc. Jobs
LiveAtMetro*WorkAtMetro	0.023***	-0.015***	-0.029***	0.067***
	[0.007]	[0.003]	[0.003]	[0.004]
WorkAtMetro	-0.016**	0.008***	-0.014***	-0.011***
	[0.002]	[0.001]	[0.001]	[0.001]
LiveAtMetro	-0.006***	-0.003***	-0.004***	0.001
	[0.001]	[0.001]	[0.001]	[0.001]
constant	0.522***	0.144***	0.190***	0.188***
	[0.000]	[0.000]	[0.000]	[0.000]
Observations	139,588,875	139,588,875	139,588,875	139,588,875
R-Squared	0.823	0.666	0.697	0.762

The unit of observation is all origin-destination pairs with at least one origin or one destination in Los Angeles County. Robust standard errors clustered by origin-destination pairs. *, **, and *** denote 0.1, 0.05, and 0.01 significance levels.